Simpson's Paradox — Corpus Summary

A digital mirror of Simpson's Paradox — the paradox falls out of the DAG as a derived fact, never modeled directly.

Generated 2026-06-29 from live Postgres views (vw_model_summary, vw_conclusions, vw_discovery_*, vw_invariant_checks). This is a narrative export — not the full rulebook.

Corpus snapshot

Studies witnessed: 96
Sign-flips (types A/B): 18
Screening tiers: DANGER 18 · CAUTION 14 · SAFE 64
Distortion taxonomy: A 7 · B 11 · C+ 8 · C" 6 · D 64
Avg signal purity: 0.787
Latent Type-D (SAFE but sweep-flip): 45 of 64 (70.3%)
LatentTypeD=45/64; SignFlipMaxPurity=0.4827404019389509953378670856050213579961; stableD=0.10212951419970321714 latentD=0.02224581026688020447

Pre-registered discovery (loop-61)

H-domain-dist — PASS

Epidemiology studies carry higher mean AllocationDistortion than education studies.

Expected: EpidemiologyAvgDistortion > EducationAvgDistortion
Observed: epi=0.0494547186001599619932270970057990291237 edu=0.0040904227907776784820219330660828745932

H-econ-zero — PASS

Economics-domain studies produce zero sign-flips in 2×K encoding.

Expected: EconomicsSignFlipCount = 0
Observed: flips=0

H-latent-d — PASS

More than half of Type-D (SAFE) studies would flip their pooled winner under counterfactual allocation reweighting.

Expected: LatentTypeDFraction > 0.5
Observed: fraction=0.70312500000000000000

H-purity — PASS

Every sign-flip study has SignalPurity strictly below 0.5.

Expected: SignFlipSignalPurityMax < 0.5
Observed: maxPurity=0.4827404019389509953378670856050213579961

H-small-effect — PASS

Allocation-stable Type-D studies carry larger observed pooled gaps than latent Type-D studies.

Expected: AvgPooledGapStableD > AvgPooledGapLatentD
Observed: stable=0.10212951419970321714 latent=0.02224581026688020447

Witnessed conclusions (17)

conc-01-paradox-is-derived · domain

Simpson's Paradox is a derived fact from the DAG, not a modeled entity

IsReversal (and later IsSignFlip) falls out as a calculated field on TreatmentRankings, derived from aggregated CaseCell counts. No 'ReversalDetection' entity was ever needed.

Witnessed in loop: [loop-04 (b96e0fe, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]
Protecting invariants: 2

conc-02-reversal-from-allocation · domain

The reversal in kidney-1986 is entirely an allocation artifact — same rates with balanced allocation produce no reversal

kidney-balanced uses identical stratum success rates to kidney-1986 but with equal allocation (175 cases per cell). IsReversal=FALSE, AllocationDistortion=0. The paradox is algebraically isolated as a property of allocation, not of treatment efficacy.

Witnessed in loop: [loop-14 (49f01e7, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]
Protecting invariants: 3

conc-03-binary-excludes-partial · instrument

The unanimity criterion (IsReversal) incorrectly excludes Berkeley 1973 despite its being a Simpson-type paradox

Berkeley: WSGSUM=" 0.0642, SignedPooledGap=+0.1292, IsSignFlip=TRUE, AllocationDistortion=0.1934 — greater distortion than kidney-1986. But IsReversal=FALSE because dept-C marginally favors male. The unanimity criterion is a strict subtype, not the primary definition.

Witnessed in loop: [loop-17 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]
Protecting invariants: 4

conc-04-type-c-exists · instrument

Allocation can compress the pooled signal without flipping it (type-C): a class invisible to IsReversal and IsSignFlip but visible to AllocationDistortion

compressed-synthetic: A wins every stratum and wins pooled, but the pooled margin is 6pp rather than the equal-weight 10pp. AllocationDistortion=0.0400, IsSignFlip=FALSE. The pooled winner is correct; the effect size is wrong by 4pp.

Witnessed in loop: [loop-19 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]
Protecting invariants: 4

conc-05-real-data-generalizes · instrument

Two real published studies (reintjes-2000, radelet-1981) hydrate as type-A full reversals without schema changes

Both studies entered via 05b-customize-data.sql. Witnessed: reintjes AllocationDistortion=0.0516, DistortionType=A; radelet AllocationDistortion=0.0684, DistortionType=A. No DAG or schema change was needed — the hydration contract generalizes.

Witnessed in loop: [loop-20b (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]

conc-06-policy-derivable · instrument

PolicyImplication is domain-neutral: the same formula produces researcher actions across medicine, law, and education

All five current studies (kidney-1986, berkeley-1973, compressed-synthetic, kidney-balanced, balanced-synthetic) plus two real studies produce unambiguous PolicyImplication values from DistortionType alone. No domain-specific knowledge is encoded in the formula.

Witnessed in loop: [loop-21 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]

conc-07-model-self-portrait · instrument

ModelSummary witnesses the instrument's own epistemic coverage as a first-class derived fact

ModelSummary aggregates ReversalCount, ExplainedCount, TypeA/B/C/D counts, and AvgAllocationDistortion across all studies. The model describes its own completeness in a single row.

Witnessed in loop: [loop-10 (b96e0fe, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]

conc-08-type-c-real · open question

Does the published literature contain real type-C studies (compression without sign flip)?

Witnessed in loop-24: hannan-1994 (CABG surgery), titanic-1912, melanoma-altman-1991, and coffee-tverdal-2020 all hydrate as DistortionType=C. AllocationDistortion>0.01, IsSignFlip=FALSE — the correct pooled direction but compressed magnitude. The type-C cell is not a synthetic pathology.

Witnessed in loop: [loop-24 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]()

conc-09-causal-vs-geometric · open question

AllocationDistortion is a geometric measure; causal interpretation requires external domain knowledge, gated by AdjustmentAppropriate

Berkeley has the highest AllocationDistortion (0.1934) but CausalRole=contested (mediator vs confounder), proving that geometric severity and causal confounding are not the same thing. The instrument is deliberately geometric: it classifies distortion type and flags the corrected winner from allocation arithmetic alone. Whether it is safe to act on that correction as a causal claim is answered by AdjustmentAppropriate — which gates on ConditioningRisk and CausalClaimStatus. The two layers are complementary, not …

Witnessed in loop: [loop-30 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]()
Protecting invariants: 1

conc-10-instrument-specifiable · methodology

The full instrument can be specified precisely enough for any researcher to apply it to a new study

Witnessed in loop-26: InstrumentSpec table enumerates 5 input fields, 10 derived coordinates, 2 classification rules, and 3 adapter-requirement rows. Machine-verifiable against the full 90+ study corpus.

Witnessed in loop: [loop-26 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]()

conc-11-reversal-recovery · instrument

The model can produce the allocation-corrected winner directly from existing DAG fields — no new data required

CorrectedGap = WeightedStratumGapSum (already derived in loop-12/17). CorrectedWinner is a sign-read on CorrectedGap. CorrectedVsPooledAgreement is TRUE for Type-D studies and FALSE for Type-A/B studies. For kidney-1986: CorrectedGap=+0.0537, CorrectedWinner=A, CorrectedVsPooledAgreement=FALSE (pooled said B; corrected says A). The reversal recovery is free — it costs zero new data.

Witnessed in loop: [loop-27 (ca31095, Sat Jun 27 2026 00:00:00 GMT-0500 (Central Daylight Time))]()
Protecting invariants: 3

conc-12-signal-purity-theorem · theorem

SignalPurity < 0.5 is a necessary condition for allocation-driven reversal

Witnessed across all 9 reversal studies (type A + B) in the 90+ study corpus: every study with AllocationDirection='reversal' has SignalPurity < 0.5. The algebraic proof: reversal requires SignedPooledGap and CorrectedGap to have opposite signs, which requires AllocationDistortion > |CorrectedGap|, which is exactly SignalPurity < 0.5. inv-signal-purity-sign-flip enforces this as a DAG invariant. AvgSignalPurityReversal "H 0.35 vs AvgSignalPurityNonReversal "H 0.65 — gap of ~0.30.

Witnessed in loop: [loop-44 (5a14dcb, Sun Jun 28 2026 00:00:00 GMT-0500 (Central Daylight Time))]()

conc-13-cplus-cminus-opposite-errors · taxonomy

C+ (amplification) and C- (compression) are epistemically opposite errors with opposite clinical implications

titanic-1912 (C+, DistortionRatio=1.359): allocation inflates first-class survival advantage from 27.5pp corrected to 37.4pp pooled — overconfidence. melanoma-altman-1991 (C+, DistortionRatio=2.087): allocation doubles apparent female survival advantage — severe overconfidence. hannan-1994 (C-, DistortionRatio=0.314): allocation compresses CABG advantage from 8.9pp corrected to 2.8pp pooled — underconfidence, risk of premature treatment abandonment. The prior type-C category hid this distinction. Now separated b…

Witnessed in loop: [loop-43 (5a14dcb, Sun Jun 28 2026 00:00:00 GMT-0500 (Central Daylight Time))]()

conc-14-scale-is-discovery-path · methodology

Cross-domain SignalPurity correlations require a pre-registered unbiased corpus of 50+ studies to distinguish pattern from selection artifact

Partial domain patterns witnessed across 90+ studies (loop-61): epidemiology avg distortion 0.050 vs education 0.004; legal/sports/epidemiology flip rates ~27% vs economics 0%. Full causal domain!’DistortionType mapping still requires pre-registered expansion beyond convenience sample, but the instrument now supports DiscoveryHypotheses/Findings for ongoing corpus-scale tests.

Witnessed in loop: [loop-61 (7b3938e, Mon Jun 29 2026 00:00:00 GMT-0500 (Central Daylight Time))]

conc-15-latent-flip-potential · instrument

Type-D at observed allocation does not imply allocation-stable — 70% of SAFE studies flip under counterfactual reweighting

LatentTypeDCount=45, TypeDCount=64, LatentTypeDFraction=0.703. LatentFlipPotential=TRUE when DistortionType=D AND PooledGapCrossesZero=TRUE. Examples: uc-systemwide-admissions (Type D, sweep range 0.56), recsys-offline-eval (Type D, sweep range 0.56). The ScreeningTier SAFE label describes observed allocation only; sweep reveals latent flip potential.

Witnessed in loop: [loop-61 (7b3938e, Mon Jun 29 2026 00:00:00 GMT-0500 (Central Daylight Time))]
Protecting invariants: 2

conc-16-small-effect-fragile · theorem

Large observed pooled gaps (>0.10) predict allocation-stable Type-D; small gaps predict latent flip potential

AvgPooledGapStableD=0.102 across 19 stable Type-D studies; AvgPooledGapLatentD=0.022 across 45 latent Type-D studies. Stable studies also have lower sweep range (avg 0.072 vs 0.211). Small observed effects are allocation-fragile even when currently classified SAFE.

Witnessed in loop: [loop-61 (7b3938e, Mon Jun 29 2026 00:00:00 GMT-0500 (Central Daylight Time))]

conc-17-economics-flip-free · domain

Economics-domain 2×K encodings in this corpus produce zero sign-flips — composition studies read as Type D or C"

EconomicsSignFlipCount=0 across 8 economics studies (6 Type D, 2 Type C"). Contrast epidemiology (6 sign-flips / 22 studies, 27.3%). Composition and wage-decomposition paradoxes compress or stabilize rather than flip in this instrument's 2×K encoding.

Witnessed in loop: [loop-61 (7b3938e, Mon Jun 29 2026 00:00:00 GMT-0500 (Central Daylight Time))]

Invariant health

Critical invariant failures: [0]
inv-pooled-gap-wanders: Pooled gap should move under allocation reweighting for reversal studies. Warning-only: small-magnitude reversals may have narrow sweep range. (fail=1)

Most misleading studies (by paradox strength)

COVID-19 CFR: Italy vs China by Age (von Kügelgen 2021) — type C+, strength 0.2743, distortion 0.529, purity 3.6% (CAUTION)
Triathlon On-Time Finish: Sex by Age Cohort — type B, strength 0.1455, distortion 0.245, purity 28.9% (DANGER)
Smoking and Mortality by Age (Cochran / Surgeon General 1964) — type B, strength 0.1091, distortion 0.167, purity 25.7% (DANGER)
UC Berkeley Graduate Admissions (Bickel et al. 1975) — contested confounder — type B, strength 0.0969, distortion 0.193, purity 24.9% (DANGER)
UC Berkeley Admissions: Six Largest Departments (Bickel et al. 1975) — type B, strength 0.0944, distortion 0.184, purity 18.8% (DANGER)
Exercise and Cholesterol Control by Age Group (Framingham-style) — type D, strength 0.0800, distortion 0.008, purity 95.2% (SAFE)
Baseball Batting Averages: Lefebvre vs Fairly World Series (Day 1994 / baseball paradox canon) — type B, strength 0.0606, distortion 0.082, purity 20.9% (DANGER)
Whickham Smoking Cohort — 20-year Mortality by Smoking Status (Appleton et al. 1996) — type B, strength 0.0538, distortion 0.119, purity 26.9% (DANGER)

Repository evidence (GitHub)

Links point at the Effortless Rulebooks monorepo. Checkout a loop tag or commit to reproduce the witnessed state.

Project folder: https://github.com/EffortlessAPI/effortless-rulebooks/tree/main/rulebook-examples/simpsons-paradox
README: README.md
Rulebook SSoT: effortless-rulebook/simpsons-paradox-rulebook.json
Rulespeak narrative: rulespeak/rulespeak.md
Generated Postgres substrate: effortless-postgres/
Standalone HTML explorer: simpsons-paradox-explorer.html

Loop landing tags (replay checkpoints):

sp-loop-milestone-01-loops-01-10: conc-01-paradox-is-derived
sp-loop-milestone-03-loops-11-15: conc-02-reversal-from-allocation
sp-loop-milestone-04-loops-16-31: conc-03-binary-excludes-partial
sp-loop-milestone-04-loops-16-31: conc-04-type-c-exists
sp-loop-milestone-04-loops-16-31: conc-05-real-data-generalizes
sp-loop-milestone-04-loops-16-31: conc-06-policy-derivable
sp-loop-milestone-01-loops-01-10: conc-07-model-self-portrait
sp-loop-milestone-04-loops-16-31: conc-08-type-c-real
sp-loop-milestone-04-loops-16-31: conc-09-causal-vs-geometric
sp-loop-milestone-04-loops-16-31: conc-10-instrument-specifiable